

AWI-ESM3-4-2-veg-HR: CMIP7 CMORization status

piControl | OpenIFS 48r1 TCo319, FESOM 2.7, LPJ-GUESS | pycmor on DKRZ Levante | 14 September 2026

Where we are

The pipeline turns one model year into a complete, CMIP7-structured data set in under four hours. Against the QC checker DKRZ will publish with (cc-plugin-wcrp coordinate checks, PR #81), the latest test year has **no HIGH findings from the CMIP7 plugin on any of its 529 files**. What remains is upstream work in the checkers and tables, a review of flagged values, and running the 196 years.

529	329	403	7	6	470 GB	3.9 h
files per year	variables	branded variables	realms	grids	per year	wall time per year

What one year contains

realm	files	variables	frequencies and grids
land	183	153	3-hourly to yearly, tiles and land-use types, TCo319 (g122)
atmos	156	55	1-hourly to monthly, native TCo319 and regridded 0.28° (g129), pressure and model levels
ocean	87	61	daily to decadal, FESOM nodes (g130) and elements (g132), basin transports (g239), global means (g010)
seaIce	78	54	daily and monthly, plus hemispheric totals (g010)
landIce, aerosol, atmosChem	25	20	monthly and yearly

Compliance: from dozens of findings to none on our side

test year	HIGH findings (cf / CMIP7 plugin / aicc)
cli117	65 with that week's checkers; DKRZ's newer aicc then also flagged longitude range (299 files), grid variable names, data types and climatologies
cli119, released checkers	0 / 42 / 1: all 42 are a known TIME001 checker bug, the 1 is a typo in the data request table
cli119, checker with PR #81	9 / 0 / 1: the 9 are a cf test that contradicts CF 1.7 itself, the 1 is the table typo

Across the DKRZ review rounds (Martin Schupfner), every finding in our files has been fixed and re-measured. Along the way we reported or fixed more than a dozen issues upstream in **compliance-checker**, **cc-plugin-wcrp**, the WCRP universe and the data request, and registered our grids in the Essential Model Documentation, including the zonal-band grid g239.

Built and working

- Year-by-year submission as SLURM shard arrays (17 tiers, 59 tasks), with QC running inside every job.
- Unstructured FESOM output with CMIP7 grid variables and cell vertices, one coordinate set per grid.
- Climatologies and monthly means of daily extremes as CF climatologies; decadal means written from ten years of input by year-by-year jobs, verified against the raw model output.
- Value checks on every file against literature ranges: 307 of 529 pass, 169 sit slightly outside.

Still to do before publication

- **Values:** understand the 53 files outside plausible ranges. 49 have been flagged since August, some are ranges set too tight (ch4 and n2o are confirmed false alarms); sea-ice and ocean transport cases need a look.
- **Checkers:** the release with the coordinate checks and the registry fix is expected by the end of September. Until then 42 files fail TIME001 in the released version.
- **Open details:** two conventions for decadal means are with DKRZ; how the running climatology file is named across years needs checking; five 6-hourly variables on seven pressure levels cannot be produced for HR.

Scale of the production run

196 years (1850–2045): about **98 600 files and 92 TB**, roughly 6500 node-hours at the current rate. Write permission on the DKRZ ESGF data node is in place.